An exact obstruction to the quadratic half-line completion

Abstract

The proposed order-33 cone is a proper subcone of the full nonnegative canonical valid-slack cone. We give an all-positive rational source and an explicit rational separating functional, certified by four positive definite dual matrices. A fixed enlargement, obtained by allowing a selector quartic in the square root of the Bob variable, contains the excluded source. Its order-49 affine pencil and exact feasible matrices are supplied. The two certificates isolate the role of the additional square-root terms.

1 The two cones and the counterexample

Use the source order

$$c = (d, \lambda, n, m, C, A, D, \alpha, \beta) \in \mathbb{R}^9_+, \qquad D_u = 1 + u^2, \quad D_v = 1 + v^2.$$

The full canonical cone $\mathcal{V}_9$ consists of the sources for which the canonical numerator

$$\begin{aligned} F_c = {} & 4\lambda D_u (W + vX)^2 + 4nD_uD_vX^2 + 4m(vW - X + uvY - uZ)^2 \\ & + 4CD_v(uW - Y)^2 + 4AD_uD_vY^2 + (4D + 2d)(uW + uvX - Y - vZ)^2 \\ & + (4\alpha + 2d)D_u(vY - Z)^2 + (4\beta + 2d)D_v(X + uZ)^2 - 2dD_uD_vZ^2 \end{aligned} \tag{1}$$

is nonnegative for every real state vector and every pair of measurement angles, including their projective endpoints. Complex states are included by separating real and imaginary parts of this real symmetric quadratic form.

The state reduction gives four quadratic matrices and two completions:

$$P = \begin{pmatrix} 4mv^2 & -4mv \\ -4mv & 4m + (4\beta + 2d)D_v \end{pmatrix}, \qquad N = \begin{pmatrix} 4CD_v + 4D + 2d & (4D + 2d)v \\ (4D + 2d)v & (4D + 2d)v^2 \end{pmatrix},$$

$$E = \begin{pmatrix} 4AD_v + (4\alpha + 2d)v^2 & -(4\alpha + 2d)v \\ -(4\alpha + 2d)v & 4\alpha - 2dv^2 \end{pmatrix}, \qquad K = 4\lambda \begin{pmatrix} 1 & v \\ v & v^2 \end{pmatrix} + 4nD_v \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \tag{2}$$

$$G_J = \begin{pmatrix} P + K & J \\ J^T & N + K \end{pmatrix}, \qquad H_J = \begin{pmatrix} P + E & K - E - J \\ K - E - J^T & N + E \end{pmatrix}. \tag{3}$$

Let $\mathcal{C}_{33}$ be the cone obtained by requiring a quadratic selector $J(v) = J_0 + vJ_1 + v^2J_2$ on $v \geq 0$. Equivalently, writing $Q(v) = Q_0 + vQ_1 + v^2Q_2$ for $Q = G_J, H_J$, its four PSD blocks are

$$S_Q = \begin{pmatrix} Q_0 & B_Q \\ B_Q^T & Q_2 \end{pmatrix} \succeq 0, \qquad T_Q = Q_1 - B_Q - B_Q^T \succeq 0. \tag{4}$$

The nine source-sign blocks give total order $8 + 8 + 4 + 4 + 9 = 33$ and 44 auxiliaries. The two-LMI identity gives $\mathcal{C}_{33} \subseteq \mathcal{V}_9$. Reflection by $T = \mathrm{diag}(1, -1)$ supplies the negative half-line; homogeneous continuity supplies the projective endpoints. No restriction on state signs is made.

Theorem 1. *The all-positive rational source*

$$\boxed{\bar{c} = \left(1, 2^{-34}, 2^{-14}, 2^{-24}, \frac{1}{576}, 161, \frac{1}{2}, 76, \frac{1}{250}\right)} \tag{5}$$

belongs to $\mathcal{V}_9$ but not to $\mathcal{C}_{33}$. Moreover, the fixed quartic-in-$\sqrt{v}$ completion cone $\mathcal{C}_{\mathrm{ram}}$ defined in Section 4 satisfies

$$\boxed{\mathcal{C}_{33} \subsetneq \mathcal{C}_{\mathrm{ram}} \subseteq \mathcal{V}_9, \qquad \bar{c} \in \mathcal{C}_{\mathrm{ram}} \setminus \mathcal{C}_{33}.}$$

It has an evaluated rational affine pencil of order 49 with nine source coordinates and 148 auxiliaries.

2 The exact separating inequality

Here is the integer linear functional that separates (5):

$$\begin{aligned}\ell(c) = &-23803618d + 906393494300170\lambda + 2744379904n\\ &+ 1378590360912m + 5935589516C + 20425A\\ &+ 2600028D + 91248\alpha + 278104928\beta.\end{aligned} \tag{6}$$

Its exact value is

$$\boxed{\ell(\bar{c}) = -\frac{5417798011632274183}{9663676416000} < -560635.} \tag{7}$$

The dual matrices below prove $\ell \geq 0$ on $\mathcal{C}_{33}$.

Lemma 2 (Dual trace identity)**.** *For $\sigma = G, H$, let $U_\sigma, R_\sigma, W_\sigma$ be symmetric 4×4 matrices satisfying*

$$M_\sigma = \begin{pmatrix} U_\sigma & R_\sigma \\ R_\sigma & W_\sigma \end{pmatrix} \succeq 0, \qquad R_\sigma \succeq 0. \tag{8}$$

For each of the three pairs (U_G, U_H), (R_G, R_H), (W_G, W_H), require their upper-right 2×2 blocks to agree. These are twelve scalar equations. Put $M_{\sigma,0} = U_\sigma$, $M_{\sigma,1} = R_\sigma$, $M_{\sigma,2} = W_\sigma$. Then

$$f(c) = \sum_{k=0}^{2} [\langle M_{G,k}, G_k(c,0)\rangle + \langle M_{H,k}, H_k(c,0)\rangle] \tag{9}$$

is nonnegative on $\mathcal{C}_{33}$. Here $G_k(c,0), H_k(c,0)$ mean the coefficients with $J = 0$, and $\langle A, B\rangle = \operatorname{tr}(AB)$.

Proof. For each side, expand (4) to obtain

$$\langle M_\sigma, S_\sigma\rangle + \langle R_\sigma, T_\sigma\rangle = \langle U_\sigma, Q_0\rangle + \langle R_\sigma, Q_1\rangle + \langle W_\sigma, Q_2\rangle.$$

The unrestricted B_σ cancels because R_σ is symmetric. The coefficients J_k enter G_k, H_k through opposite off-diagonal blocks, so the twelve overlap equations cancel every entry of J_0, J_1, J_2. Consequently

$$f(c) = \langle M_G, S_G\rangle + \langle R_G, T_G\rangle + \langle M_H, S_H\rangle + \langle R_H, T_H\rangle \geq 0. \tag{10}$$

□

Appendix A displays six rational 4×4 matrices satisfying (8) and the twelve overlap equations, with

$$f(c) = 2^{-42}\ell(c). \tag{11}$$

Their four PSD blocks are in fact positive definite. The source-sign dual variables are all zero. Equations (7)–(11) therefore prove $\bar{c} \notin \mathcal{C}_{33}$ by finite rational arithmetic.

This also excludes a polynomial half-line selector of any degree. Indeed, every 2×2 principal minor of G_J gives

$$|J_{ij}(v)|^2 \leq (P + K)_{ii}(v)(N + K)_{jj}(v) = O(v^4) \qquad (v \longrightarrow +\infty).$$

Every polynomial entry must therefore have degree at most two, already excluded by the certificate above. Every selector for this source therefore depends nonpolynomially on v.

3 Validity of the source

Validity follows from the rational contact source

$$\begin{aligned}(d, \lambda, n) &= (1, 0, 0),\\ (m, C, A) &= \left(\frac{1166523000}{31661820415017653}, \frac{322803000}{186199784281}, \frac{218288946200}{1359122513}\right),\\ (D, \alpha, \beta) &= \left(\frac{90943434293}{222507771414}, \frac{248802430236}{3300726103}, \frac{205476973}{51646655494}\right),\end{aligned} \tag{12}$$

denoted c_*. Its exact contact is $(u,v,\psi)=(1/13,16,(140,-1/32,1/50,1))$. Let $\Pi_*(u,v)$ be the 4×4 state matrix of (1), and set $x=169u^2$, $y=v^2/256$. The retained determinant identity can be written

$$\det\Pi_* = 16\kappa D_u^2 D_v^2 u^2 v^2\,\mathcal{A}(x,y),\quad \kappa=\frac{365997345792000000}{80653488310452717062149398114218073953704585845 7}>0, \tag{13}$$

where

$$\begin{aligned}\mathcal{A}(x,y) &= \rho_1(x+x^{-1}-2)+\rho_2(y+y^{-1}-2)\\ &\quad+\rho_3(y/x+x+y^{-1}-3)+\rho_4(xy+x^{-1}+y^{-1}-3),\end{aligned}$$

and the positive integer coefficients are

ρ_1	126791783324459354102518032323 1
ρ_2	762274793296209851400745596627083
ρ_3	27377911873078066203947999960 19
ρ_4	791658120509994203488525245 00

Each bracket is nonnegative by the two- or three-term AM–GM inequality. To pass from determinant positivity to all-state positivity, the $Z=0$ principal block B_* satisfies

$$B_*\succeq T_*^T\,\mathrm{diag}(4m,4CD_v,4AD_uD_v)T_*\succ 0,\qquad T_*=\begin{pmatrix}v&-1&uv\\u&0&-1\\0&0&1\end{pmatrix},\quad \det T_*=u \tag{14}$$

when $u\neq 0$. All omitted terms of (1) are nonnegative on $Z=0$. Thus the scalar Schur complement and (13) give $\Pi_*\succeq 0$ for $uv\neq 0$. Polynomial continuity recovers the axes; bihomogenization in the two angle pairs recovers every projective endpoint.

Direct rational comparison gives $\bar{c}-c_*\geq 0$, with the d coordinate unchanged. Every other coordinate in (1) multiplies a nonnegative event form. Therefore

$$F_{\bar{c}}=F_{c_*}+F_{\bar{c}-c_*}\geq 0$$

for all states and tests. This proves $\bar{c}\in\mathcal{V}_9$. The unrounded contact source itself is also separated:

$$\ell(c_*)=-\frac{30372270056956512321138766804 0}{267603838100324717114473}<0. \tag{15}$$

4 One fixed enlargement that repairs the obstruction

Set $v=16s^2$ on the positive half-line. Allow a selector $J(s)=\sum_{k=0}^4 J_k s^k$. This adds the selector powers $\sqrt{v}$ and $v^{3/2}$, with auxiliary coefficients independent of the testing variable. Put

$$Z_2(s)=\begin{pmatrix}I_4\\sI_4\\s^2I_4\end{pmatrix},\qquad Z_1(s)=\begin{pmatrix}I_4\\sI_4\end{pmatrix}.$$

For $Q=G_J,H_J$, impose the identity

$$Q(16s^2)=Z_2(s)^TA_QZ_2(s)+sZ_1(s)^TB_QZ_1(s),\qquad A_Q\succeq 0,\quad B_Q\succeq 0. \tag{16}$$

Here the notation on the left means substitution in the source blocks, with $J=J(s)$. The block orders are $12,8,12,8$ and nine scalar source blocks. This defines $\mathcal{C}_{\mathrm{ram}}$ and immediately proves $\mathcal{C}_{\mathrm{ram}}\subseteq\mathcal{V}_9$ using the two-LMI identity. Matrix half-line factorization also shows that (16) captures exactly the quartic-in-s selector condition, including singular matrices. Substituting $v=16s^2$ in any quadratic half-line certificate gives one of these Grams, proving $\mathcal{C}_{33}\subseteq\mathcal{C}_{\mathrm{ram}}$.

Eliminating the coefficient equations. For a prescribed quartic $Q(s)=\sum_{k=0}^{4}Q_k s^k$, write $A=(A_{ij})_{i,j=0}^{2}$ and $B=(B_{ij})_{i,j=0}^{1}$ in 4×4 blocks. Choose B_{00},B_{11} symmetric, B_{01},A_{02} arbitrary, and K_{01},K_{12} skew-symmetric. Set

$$\begin{aligned} A_{00}&=Q_0, & A_{22}&=Q_4,\\ A_{01}&=\tfrac{1}{2}(Q_1-B_{00})+K_{01}, & A_{12}&=\tfrac{1}{2}(Q_3-B_{11})+K_{12},\\ A_{11}&=Q_2-A_{02}-A_{02}^T-B_{01}-B_{01}^T. \end{aligned} \tag{17}$$

Transpose the lower blocks. These formulas parametrize every coefficient solution of (16). They use $10+10+16+16+6+6=64$ variables per side, in addition to the 20 selector coefficients. Thus the final pencil has 148 auxiliaries and no coefficient equations. The exported 49-dimensional pencil has 411 nonzero upper-triangular rational coefficient entries, with every source and auxiliary coefficient evaluated.

Exact repair of both sources. The supplied data give rational face matrices V_i and positive definite rational matrices X_i of orders $7,3,11,7$. Their congruences give the four Gram matrices in (16) at c_*. Every unpivoted LDL pivot of X_i is greater than 2^{-18}; all 80 coefficient conditions after selector elimination are checked exactly. The full rational Grams for $\bar{c}$ are also exported, together with their 148 affine-pencil auxiliaries. Their PSD proof is the same four positive congruences plus nonnegative rational event outer products.

The extension from c_* to $\bar{c}$ is elementary. For $\delta=\bar{c}-c_*$, one has $d_\delta=0$, and $P_\delta,N_\delta,E_\delta,K_\delta$ are sums of polynomial event outer products. Taking $J_\delta=K_\delta$ gives

$$\begin{aligned} G_\delta&=\operatorname{diag}(P_\delta,N_\delta)+\begin{pmatrix}K_\delta & K_\delta\\ K_\delta & K_\delta\end{pmatrix},\\ H_\delta&=\operatorname{diag}(P_\delta,N_\delta)+\begin{pmatrix}E_\delta & -E_\delta\\ -E_\delta & E_\delta\end{pmatrix}. \end{aligned}$$

These additions lie in the same quartic-in-s family. This also proves that the entire $d=0$ source face is included, with arbitrary zero event weights. For example, the new selector at c_* has the nonzero odd coefficient

$$[s]J_{21}(s)=\frac{4449008858798426042034668223711 09}{3909264470979595000000000000000000}.$$

The explicit certificate uses the additional square-root terms. The separation proves that some function outside the quadratic span is necessary.

The fixed scalar testing family. To express the Gram certificate directly in the physical state $\psi=(W,X,Y,Z)^T$, set

$$\eta=L(u)\psi,\quad L(u)=\begin{pmatrix}u & 0 & u^2 & 0\\ 0 & u & 0 & u^2\\ u & 0 & -1 & 0\\ 0 & u & 0 & -1\end{pmatrix},\quad U=u^2,\quad D_U=\operatorname{diag}(1,1,U,U).$$

For $v\neq 0$ put $s=\sqrt{|v|/16}$ and $\eta_\sigma=\operatorname{diag}(1,\sigma,1,\sigma)\eta$, where $\sigma=\operatorname{sgn}(v)$. At $v=0$ choose $\sigma=1$. The four groups of fixed functions are

$$\boxed{Z_2(s)D_U\eta_\sigma,\quad \sqrt{s}\,Z_1(s)D_U\eta_\sigma,\quad uZ_2(s)\eta_\sigma,\quad u\sqrt{s}\,Z_1(s)\eta_\sigma.} \tag{18}$$

They have sizes $12,8,12,8$ and their block Gram form equals $u^2D_uF_c$. The Bob factors are exactly $\{1,s,s^2,\sqrt{s},s^{3/2}\}$, so the added Gram factors are fixed quarter powers of $|v|$. Source and testing variables remain independent. Dividing every function by $D_u^{3/2}\sqrt{D_v}$ represents

$$\frac{u^2}{D_u}\,\frac{F_c}{D_uD_v}. \tag{19}$$

This division involves only testing variables. Here is a literal compact algebraic cover, including every endpoint. Take

$$a_0^2+a_1^2=1,\quad a_0\geq 0,\qquad \rho^2+\tau^2=1,\quad \rho,\tau\geq 0,\qquad \epsilon\in\{-1,1\},$$

and put

$$\kappa=\sqrt{\tau^8+256\rho^8},\quad b=\frac{(\tau^4,16\epsilon\rho^4)}{\kappa},\quad g=\frac{(\tau^4,\rho^2\tau^2,\rho^4)}{\kappa},\quad r=\frac{(\rho\tau^3,\rho^3\tau)}{\kappa}.$$

These are source-independent algebraic functions with $\kappa^2 \geq \tau^8+\rho^8 \geq 1/8$. In the interiors, $u = a_1/a_0$, $v = 16\epsilon(\rho/\tau)^4$. Define the homogeneous cubic vectors

$$P_G = \begin{pmatrix} a_0^2 a_1 W + a_0 a_1^2 Y \\ \epsilon(a_0^2 a_1 X + a_0 a_1^2 Z) \\ a_1^3 W - a_0 a_1^2 Y \\ \epsilon(a_1^3 X - a_0 a_1^2 Z) \end{pmatrix}, \qquad P_H = \begin{pmatrix} a_0 a_1^2 W + a_1^3 Y \\ \epsilon(a_0 a_1^2 X + a_1^3 Z) \\ a_0 a_1^2 W - a_0^2 a_1 Y \\ \epsilon(a_0 a_1^2 X - a_0^2 a_1 Z) \end{pmatrix}.$$

The forty functions on this closed cover are

$$g \otimes P_G, \quad r \otimes P_G, \quad g \otimes P_H, \quad r \otimes P_H.$$

Their Gram sum is $a_1^2 f_c(a, b; \psi)$, where f_c is the bihomogeneous normalized physical slack. This follows from (19) on the interior and is an algebraic identity everywhere. Each function has absolute value less than 3 when $\|\psi\| = 1$. At Bob's endpoints, $(g, r) = ((1, 0, 0), 0)$ or $((0, 0, 1/16), 0)$; the Alice endpoints are literal polynomial substitutions in P_G, P_H. The multiplier a_1^2 is positive on a dense testing set. Consequently every state sign and every endpoint is retained in the validity converse.

Exchanging Alice and Bob gives an all-positive valid source outside $\mathcal{C}_{\text{ram}}$. Thus $\mathcal{C}_{\text{ram}} \subsetneq \mathcal{V}_9$. Increasing the polynomial degree in s cannot repair this exclusion: the PSD principal-minor bounds force every polynomial selector to have degree at most four.

A Six rational matrices proving the separation

Put $q = 2^{30}$. The following are the hatted moment matrices. The lower triangles are displayed as well so that there is no storage convention ambiguity.

$$q\widehat{U}_G = \begin{pmatrix} 3376576 & -4595 & 8890485/2^{20} & -141059/2^{14} \\ -4595 & 82702 & 374101/2^7 & -2966 \\ 8890485/2^{20} & 374101/2^7 & 48657 & -1435456 \\ -141059/2^{14} & -2966 & -1435456 & 159900862 \end{pmatrix},$$

$$q\widehat{R}_G = \begin{pmatrix} 2570 & 373 & 100461333/2^{23} & -796959/2^{16} \\ 373 & 5619 & 132105/2^5 & -4192 \\ 100461333/2^{23} & 132105/2^5 & 4654 & -50824 \\ -796959/2^{16} & -4192 & -50824 & 1442242 \end{pmatrix},$$

$$q\widehat{W}_G = \begin{pmatrix} 20 & 26 & 156439171/2^{23} & -18103/2^{10} \\ 26 & 822 & 92585/2^7 & -4442 \\ 156439171/2^{23} & 92585/2^7 & 697 & -4948 \\ -18103/2^{10} & -4442 & -4948 & 53164 \end{pmatrix},$$

$$q\widehat{U}_H = \begin{pmatrix} 142545776 & 374885 & 142247760 & -564236 \\ 374885 & 1002 & 374101 & -1483 \\ 142247760 & 374101 & 141950400 & -563057 \\ -564236 & -1483 & -563057 & 2251 \end{pmatrix},$$

$$q\widehat{R}_H = \begin{pmatrix} 201343592 & 529527 & 200922666 & -796959 \\ 529527 & 1401 & 528420 & -2096 \\ 200922666 & 528420 & 200502635 & -795292 \\ -796959 & -2096 & -795292 & 3163 \end{pmatrix},$$

$$q\widehat{W}_H = \begin{pmatrix} 313472891 & 94149 & 312878342 & -1158592 \\ 94149 & 16684 & 92585 & -2221 \\ 312878342 & 92585 & 312285067 & -1156239 \\ -1158592 & -2221 & -1156239 & 4509 \end{pmatrix}.$$

Set

$$\Delta_G = \text{diag}(128, 1/32, 1/2, 1/32), \qquad \Delta_H = \text{diag}(1/512, 1/16, 1/512, 1/32).$$

The moments used in Lemma 2 are

$$U_\sigma = \Delta_\sigma \widehat{U}_\sigma \Delta_\sigma, \quad R_\sigma = 16\Delta_\sigma \widehat{R}_\sigma \Delta_\sigma, \quad W_\sigma = 256\Delta_\sigma \widehat{W}_\sigma \Delta_\sigma.$$

The twelve upper-right-block equalities follow entry by entry. For positive definiteness, apply the rational recurrence

$$p_j = A_{jj} - \sum_{k<j} L_{jk}^2 p_k, \qquad L_{ij} = \frac{A_{ij} - \sum_{k<j} L_{ik} L_{jk} p_k}{p_j}, \quad L_{jj} = 1.$$

For $\widehat{M}_\sigma = \left(\begin{smallmatrix} \widehat{U}_\sigma & \widehat{R}_\sigma \\ \widehat{R}_\sigma & \widehat{W}_\sigma \end{smallmatrix}\right)$, the exact floors of $2^{30} p_j$ are

$\widehat{M}_G$	$(3376576, 82695, 48553, 117468710, 16, 14, 14, 31)$
$\widehat{R}_G$	$(2570, 5564, 1593, 10646)$
$\widehat{M}_H$	$(142545776, 16, 32, 17, 29078313, 12, 24, 11)$
$\widehat{R}_H$	$(201343592, 8, 15, 8)$.

The first scaled pivot in every row equals the listed integer; each later pivot lies strictly between its listed integer and that integer plus one. In particular, every pivot is greater than 2^{-27}. Finally, direct coefficient pairing with (2) gives (11). These six small matrices, the stated recurrence, and one rational substitution prove the separation.

B Closedness and strict feasibility of the ramified cone

An integral bound and closedness. For indices starting at zero, put

$$(H_3)_{ij} = \frac{1}{i+j+1} \quad (0 \le i, j \le 2), \qquad (H_2')_{ij} = \frac{1}{i+j+2} \quad (0 \le i, j \le 1),$$

and $W = \operatorname{diag}(H_3 \otimes I_4, H_2' \otimes I_4, H_3 \otimes I_4, H_2' \otimes I_4)$. This is positive definite: its factors are the moment matrices of $1, s, s^2$ and of $1, s$ with weights 1 and s on $[0,1]$. Their inverse traces are 381 and 54, respectively, so $W \succeq I_{40}/381$. The completion terms cancel in the trace, giving the exact coefficient identity

$$\langle W, \operatorname{diag}(A_G, B_G, A_H, B_H)\rangle = \int_0^1 \operatorname{Tr}(G_J(16s^2) + H_J(16s^2))\, ds = 8 \int_0^1 (1 + 256 s^4)\, ds \sum_{k=0}^{8} c_k = \frac{2088}{5} \sum_{k=0}^{8} c_k. \quad (20)$$

Thus W annihilates all 148 auxiliary coefficient matrices. If all four Gram blocks are positive semidefinite, bounded source coordinates bound their total trace and hence every entry. The auxiliary variables are recovered by fixed linear operations: the J_k are the upper-right 2×2 blocks of the coefficients of $G_J(16s^2)$; $B_{00}, B_{11}, B_{01}, A_{02}$ are Gram subblocks; and K_{01}, K_{12} are the skew parts of A_{01}, A_{12}. Bounded sources therefore admit bounded feasible auxiliary witnesses. Taking a subsequence proves that the projected source cone is closed. For a strictly positive weight on the whole pencil, use $\widetilde{W} = \operatorname{diag}(I_9, W)$; it still annihilates the auxiliaries, and its nine source contractions are all $2093/5$.

A strict primal point. The rational order-33 certificate in the accompanying strict-feasibility proof has $c^0 = (1/100, 1, 1, 1, 1, 1, 1, 1, 1)$ and $J(v) = K(v)$. For either of its completion polynomials $Q(v) = Q_0 + vQ_1 + v^2 Q_2$, write its established certificate as

$$S = \begin{pmatrix} Q_0 & B \\ B^T & Q_2 \end{pmatrix}, \qquad T = Q_1 - B - B^T = I_4/50.$$

The two old bounds are $S_G \succeq 19 I_8/25$ and $S_H \succeq 179 I_8/300$. Consequently

$$A^0 = \begin{pmatrix} Q_0 & 0 & 16B \\ 0 & 16T & 0 \\ 16B^T & 0 & 256 Q_2 \end{pmatrix} \succeq \frac{8}{25} I_{12}.$$

Set $B^{\mathrm{new}} = I_8/100$, and replace the 01 and 12 blocks of A^0, and their transposes, by $-I_4/200$. These changes cancel the additional coefficients $sI_4/100 + s^3 I_4/100$ from $sZ_1^T B^{\mathrm{new}} Z_1$, preserving the exact polynomial $Q(16s^2)$. The symmetric perturbation has maximum absolute row sum $1/100$. Hence $A^{\mathrm{new}} \succeq 31 I_{12}/100$ for both $Q = G, H$. The full order-49 pencil is therefore at least $I_{49}/100$.

In the actual 148-coordinate parametrization, this point has

$$(J_0, J_1, J_2, J_3, J_4) = \left(4I_2, 0, \begin{pmatrix} 0 & 64 \\ 64 & 0 \end{pmatrix}, 0, \begin{pmatrix} 0 & 0 \\ 0 & 2048 \end{pmatrix}\right),$$

and, on each side, $B_{00} = B_{11} = I_4/100$, $B_{01} = 0$, $A_{02} = 16B$, $K_{01} = K_{12} = 0$. The literal auxiliary vector and the four Gram blocks are supplied in the accompanying rational data.